

Week 2 – Identifying Data Requirements

E-Commerce Management System

1. Objective: Identify the data that must be stored in the e-commerce database. Each entity contains attributes describing it, and a Primary Key (PK) uniquely identifies each record.

2. Entities, Attributes and Primary Keys

Entity	Attributes	Primary Key
Customer	Customer_ID, First_Name, Last_Name, Email, Phone, Password, Address, City, Pincode	Customer_ID
Product	Product_ID, Product_Name, Description, Price, Stock_Quantity, Category_ID, Supplier_ID	Product_ID
Category	Category_ID, Category_Name, Description	Category_ID
Supplier	Supplier_ID, Supplier_Name, Email, Phone, Address	Supplier_ID
Order	Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address	Order_ID
Payment	Payment_ID, Order_ID, Payment_Date, Amount, Payment_Method, Payment_Status, Transaction_ID	Payment_ID
Shipment	Shipment_ID, Order_ID, Shipment_Date, Delivery_Date, Tracking_Number, Shipping_Status, Delivery_Address	Shipment_ID
Review	Review_ID, Customer_ID, Product_ID, Rating, Comment, Review_Date	Review_ID

3. Foreign Keys and Relationships

Product: Category_ID → Category(Category_ID); Supplier_ID → Supplier(Supplier_ID)

Order: Customer_ID → Customer(Customer_ID)

Payment: Order_ID → Order(Order_ID)

Shipment: Order_ID → Order(Order_ID)

Review: Customer_ID → Customer(Customer_ID); Product_ID → Product(Product_ID)

4. Entity Relationships

Customer places **Order** → Order has **Payment** and **Shipment**.

Product belongs to **Category** and is supplied by **Supplier**.

Customer writes **Review** for **Product**.

5. Conclusion

The identified entities, attributes, primary keys, and foreign keys form the basic data requirements of the E-Commerce Management System. They will be used to create the ER diagram and implement the database in MySQL.